Covariance Localization Is Schur Damping, and Its Optimal Strength Is a Sampling-Error Correction

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Abstract

Ensemble Kalman filters damp the cross-covariances of a small-ensemble forecast covariance by an elementwise product with a distance taper. We observe that a taper which is constant on blocks is the Schur-complementary damping of [1]: the localized covariance is positive semidefinite, its conditional covariance of the unobserved block is the damped Schur complement $A - \gamma B D^{-1} B^\top$, and the localized gain is $\sqrt{\gamma}$ times the optimal gain. Under sampling error in the cross-covariance the expected analysis error is an exact quadratic in $\sqrt{\gamma}$, so the optimal coupling has the closed form of a signal-to-noise shrinkage of the gain, strictly interior at every positive noise level when the signal and sampling-error terms are both positive.

1 Setting

Partition a state into an unobserved block and an observed block, with forecast error covariance $\Sigma = \left(\begin{smallmatrix} A & B \\ B^\top & D \end{smallmatrix}\right) \succ 0$. Observations of the second block carry noise of covariance R, so the innovation has covariance $S = D + R$. The optimal gain for the unobserved block is $K_\star = BS^{-1}$ and its analysis error covariance is $A - BS^{-1}B^\top$. With $R = 0$ this is the Schur complement of A in Σ.

An ensemble Kalman filter [2] replaces Σ by a sample covariance from a small ensemble. Because the sample cross-covariances are noisy, the filter localizes: it multiplies the sample covariance elementwise by a taper C that decays with distance [3, 4, 5]. The product $\Sigma \circ C$ is positive semidefinite whenever C is, by the Schur product theorem. The name is Schur's, but the operation is the elementwise product, and the strength of localization is chosen empirically.

2 A block-constant taper is Schur damping

Fix $\gamma \in [0, 1]$ and let C_γ equal 1 on the two diagonal blocks and $\sqrt{\gamma}$ on the two off-diagonal blocks.

Proposition 1 (Block taper)**.** *C_γ is positive semidefinite, so $\Sigma \circ C_\gamma \succeq 0$. The conditional covariance of the unobserved block in $\Sigma \circ C_\gamma$ is the damped Schur complement*

$$A - \gamma\, B D^{-1} B^\top \tag{1}$$

of [1], and the localized gain is $\sqrt{\gamma}\, K_\star$.

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Proof. $C_\gamma = \sqrt{\gamma}\,\mathbf{1}\mathbf{1}^\top + (1-\sqrt{\gamma})\,(\mathbf{1}_A\mathbf{1}_A^\top + \mathbf{1}_D\mathbf{1}_D^\top)$ is a nonnegative combination of rank-one positive semidefinite matrices. The tapered matrix has blocks A, $\sqrt{\gamma}B$ and D, so its Schur complement is $A - \sqrt{\gamma}B\,D^{-1}\sqrt{\gamma}B^\top$, and its gain is $\sqrt{\gamma}\,BS^{-1}$. □

So a block-constant localization is the coupling dial of Schur-complementary allocation. No localization is $\gamma = 1$ and complete localization is $\gamma = 0$, the ends of the bridge in [1, 6]. Distance tapers vary within blocks and are not of this form; the block case is the one in which the complement description is exact.

3 The optimal coupling under sampling error

Let the cross-covariance be estimated with error, $\widehat{B} = B + \tau E$, where E has mean zero, a law unchanged by $E \mapsto -E$, and is independent of the forecast and observation errors. Take S as known. The localized filter applies the gain $s\,\widehat{B}S^{-1}$ with $s = \sqrt{\gamma}$.

Proposition 2 (Exact expected analysis error)**.** *The expected analysis error covariance of the unobserved block is*

$$F(s) = A - (2s - s^2)\,Q + s^2\tau^2\,G, \qquad Q = BS^{-1}B^\top, \quad G = \mathbb{E}\big[ES^{-1}E^\top\big]. \tag{2}$$

Its trace is minimized at

$$s_\star = \frac{\operatorname{tr} Q}{\operatorname{tr} Q + \tau^2 \operatorname{tr} G}, \qquad \gamma_\star = s_\star^2. \tag{3}$$

If $\operatorname{tr} G > 0$ *then* $\gamma_\star < 1$ *for every* $\tau > 0$*, and if also* $\operatorname{tr} Q > 0$ *then* $\gamma_\star > 0$*, so the optimum is strictly interior. At* $\tau = 0$ *it is* $\gamma_\star = 1$*, and with* $\operatorname{tr} Q = 0$ *it is* $\gamma_\star = 0$*.*

Proof. With gain K the analysis error covariance is $A - KB^\top - BK^\top + KSK^\top$. Put $K = s(B + \tau E)S^{-1}$ and take expectations. The terms linear in E vanish by sign symmetry, the cross terms vanish by independence, and $\mathbb{E}[KSK^\top] = s^2(Q + \tau^2 G)$. The trace of (2) is a convex quadratic in s with minimizer (3); the boundary cases are read off that formula. □

The optimal gain is the sample gain shrunk by $s_\star$, the ratio of the variance the true gain removes to that quantity plus the variance the sampling error adds. This is a sampling-error correction in the sense of [9], applied uniformly to a block rather than entry by entry. The Schur coupling is its square. With independent standard normal entries in E, $G = \operatorname{tr}(S^{-1})\,I$.

Remark 1 (Contrast with the portfolio bridge)*.* On the portfolio bridge the full-coupling end can be the out-of-sample optimum or not, according to the sign of one derivative there [6, 7]. Here the optimum is interior at every positive noise level once both terms of (3) are positive. The reason is that the analysis mean is linear in the gain and there is no budget constraint, so the objective is an exact quadratic in s, and the companion vector of the allocation problem has no counterpart.

Remark 2 (Scope)*.* Taking S as known and E independent of the errors are simplifications. In an ensemble filter the sampling errors of $\widehat{B}$ and $\widehat{D}$ come from the same ensemble and are correlated with each other and with the forecast errors. Proposition 1 is exact regardless. Proposition 2 is the leading-order model of the trade-off, and the quantities in (3) are estimable from the ensemble. We have not found the complement formulation in the data assimilation literature; the nearest relatives are the sampling-error corrections of [8, 9].

References

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